CHAPER 2: Distributed Computer System DCS Definition

1- Autonomous Computer System

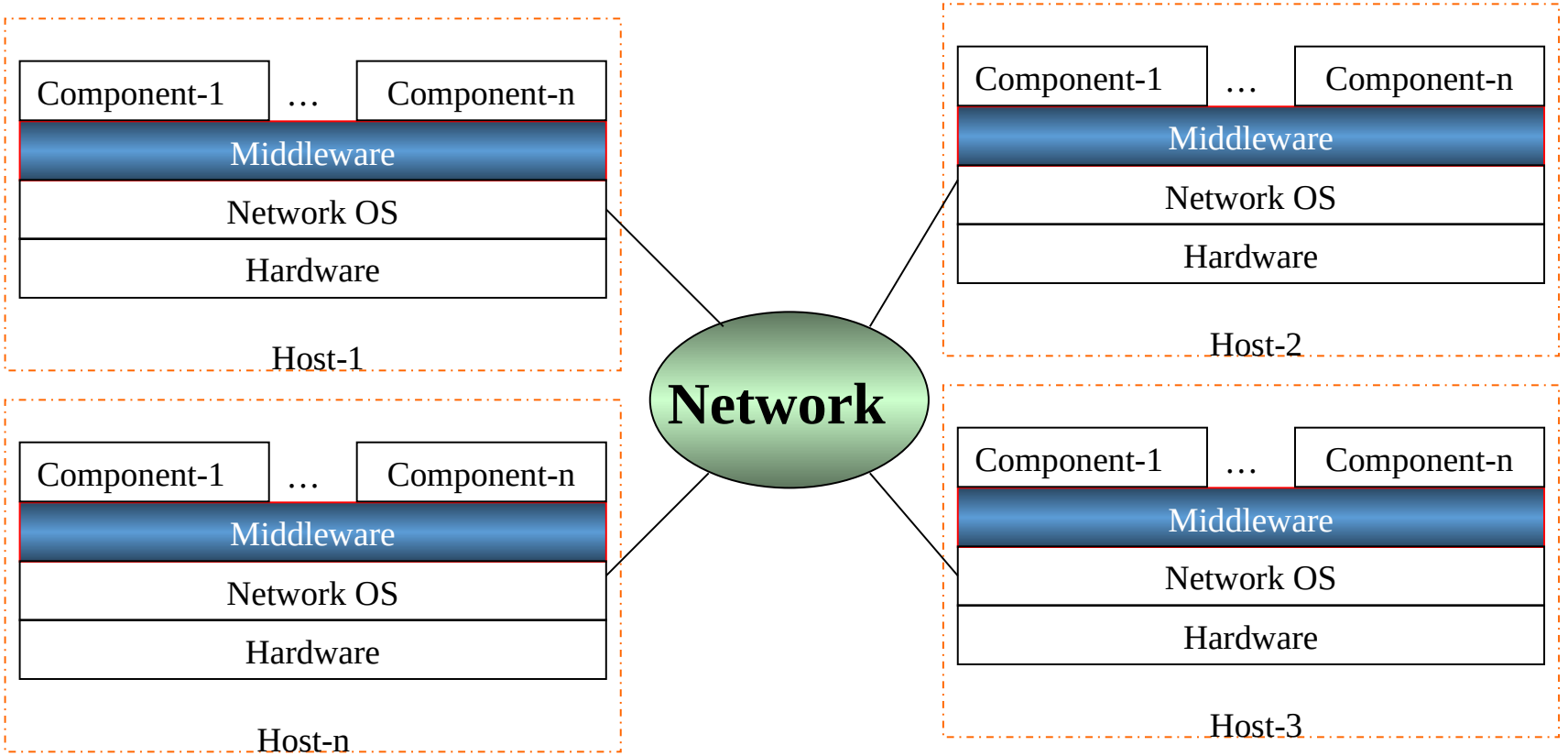
A Distributed Computer System is a system with:

1.Group of autonomous system

2.Geographically dispersed

3.Connected via a network

4.Having common/shared applications



2- Geography Dispersed

3- connected Via Networks

4- having common applications

Components of Distributed Software:
Distributed operating systems
Middleware
Distributed applications

DCSs Objectives:

Advantages of DISTRIBUTION	Advantages of Centralization
1) Technological changes	1) Economy of scale
2) User needs	2) More powerful capabilities
3) Modularity and simpler software	3) Operating costs
4) Flexibility and Extension	4) Staff Satisfaction
5) Availability and integrity	5) Local programming and development
6) Performance	6) Communication subsystem
7) Local control	7) Stretching the state of the art

The technologies that move the computer computing systems from centralized to distributed

Starting in the mid-1980s, **four advances in technology** began to change from centralized to distributed computer systems:



Microcomputer and microprocessor technology

Evolution of Local Area Network Technology

PCs Technology

Need for multimedia for decision making

1. Advances in Microcomputer and microprocessor technology

The first was the development of powerful microprocessors.

Initially, these were 8-bit machines, but soon **16-, 32-, and 64-bit CPUs** became common. With multicore CPUs, we now are facing the challenge of adapting and developing programs to **exploit clustering and parallelism**.

Current generation of machines have the computing power of the mainframes deployed 30 or 40 years ago, but for 1/1000th of the price or less.

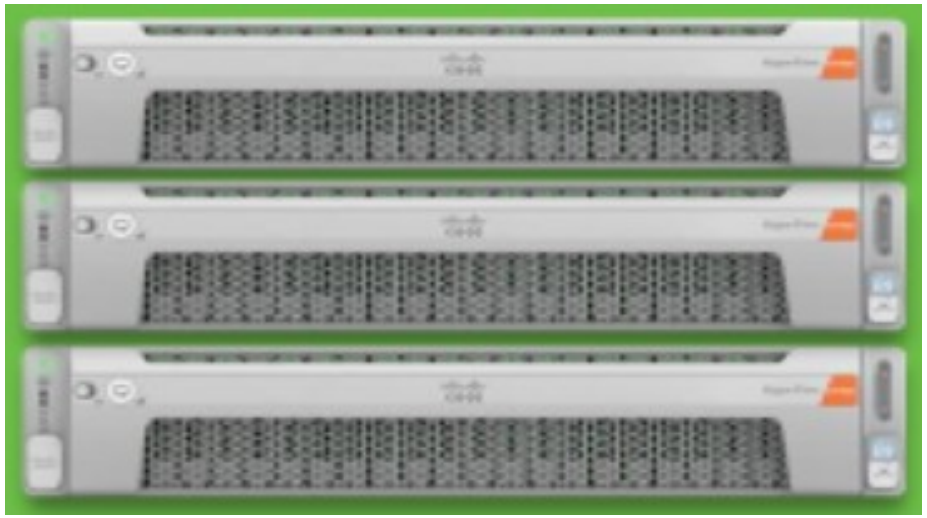
Ex1: HP PC: Windows 11, Up to **12 Core i9 processor**, 14 threads, 12MB L3CACHE, Up to 128 GB DDR5 RAM, and 2.5 TB STORAGE

Ex2: Hyper-converged Infrastructure HCI is a SDH (hardware/software-defined IT infrastructure) that virtualizes all of the elements of conventional "hardware-defined by software" systems.
HCI includes clustering and virtualized computing…

Hyperflex Hyper Converted Infrastructure HCI

Hyper-converged infrastructure HCI is a hardware/software-defined IT infrastructure that virtualizes all of the elements of conventional "hardware-defined" systems.

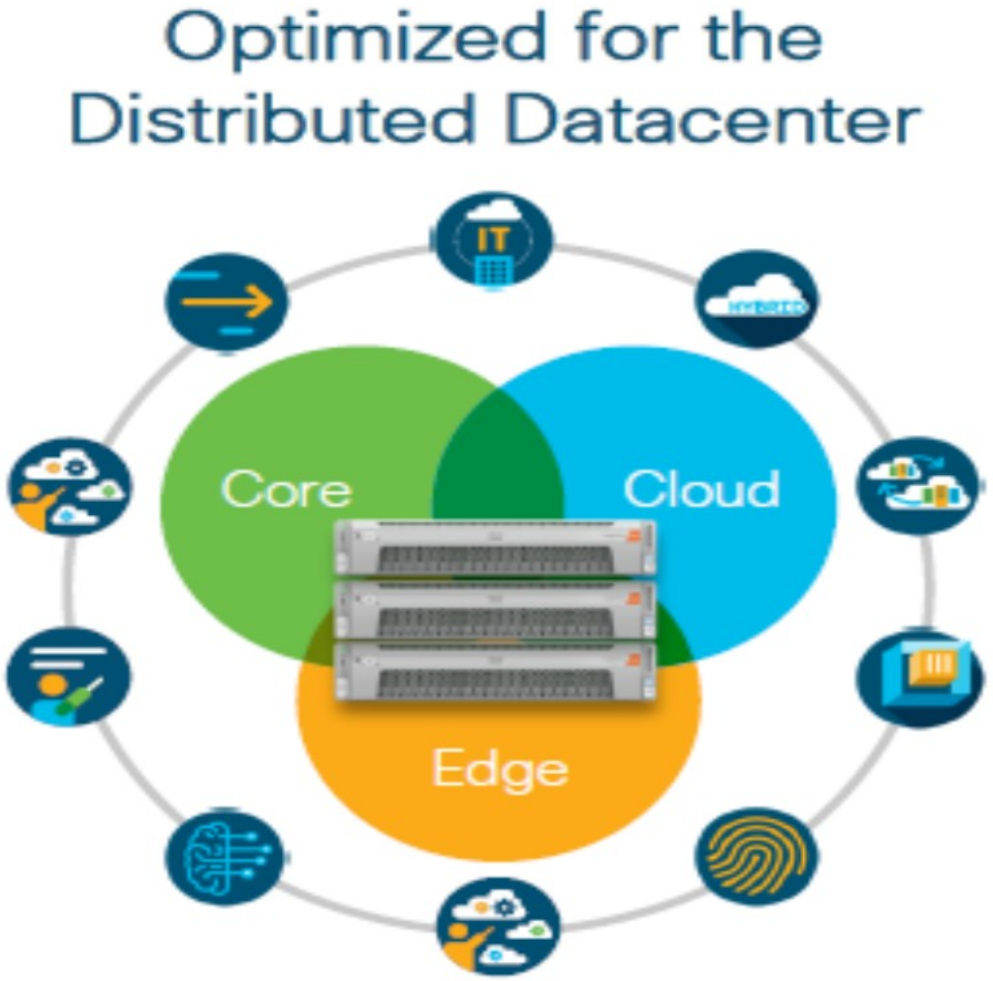
HCI includes, at a minimum, clustered of at least 3 processors with virtualized computing…



▪**Hyper Converged Infrastructure.**

-HCI Nodes All-Flash HX240C.

- -16 CPU
- -512 cores, 1024 threats.
- -3 TB Memory
- -200 TB SSD Storage



1. Advances in Microcomputer and microprocessor technology

The categories of computers and their uses based on differences in size, speed, processing capabilities, and price

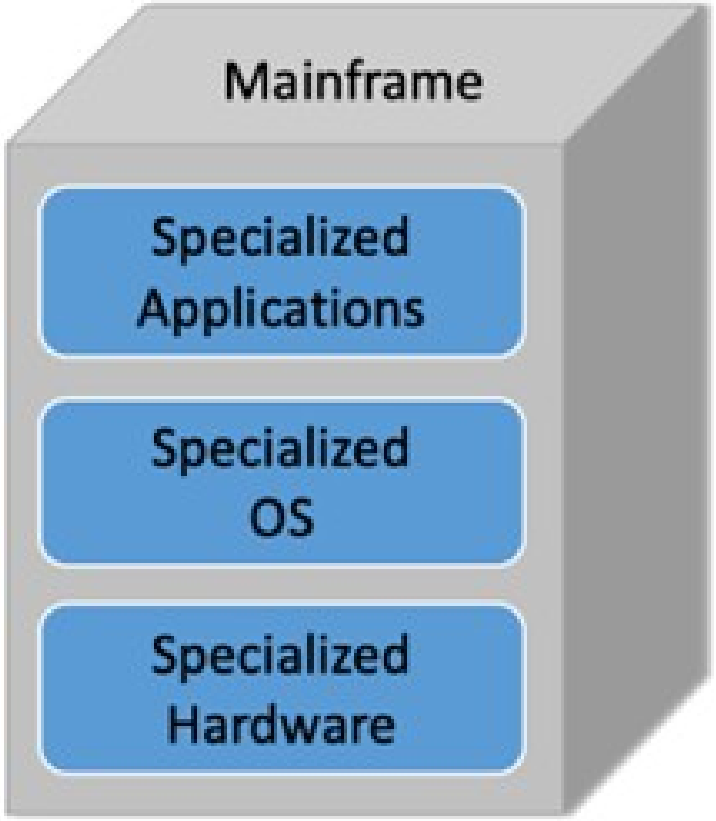
personal computers	handheld computers	Internet appliances	mid-range servers	mainframes	supercomputers
A PC can perform all of its input, processing, output, and storage activities by itself. PCs include desktop computers and notebook computers.	A **handheld computer** is a small computer that fits in hand. Handheld computers can perform specific, industry-related functions, or can be general-purpose. A PDA is a handheld computer that provides personal organizer functions, such as a calendar, appointment book, and notepad.	An Internet appliance is a computer with limited functionality whose main purpose is to connect to the Internet from home.	A **mid-range server** is more powerful and larger than a workstation computer. Users typically access a mid-range server through a personal computer or a terminal, which is a device with a monitor and a keyboard that usually has no stand-alone processing power.	A **mainframe is a large**, expensive, very powerful computer that can handle hundreds or thousands of connected users simultaneously.	A **supercomputer** is the fastest, most powerful, and most expensive category of computer.

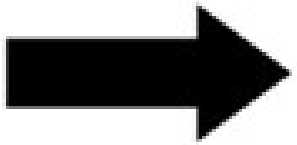


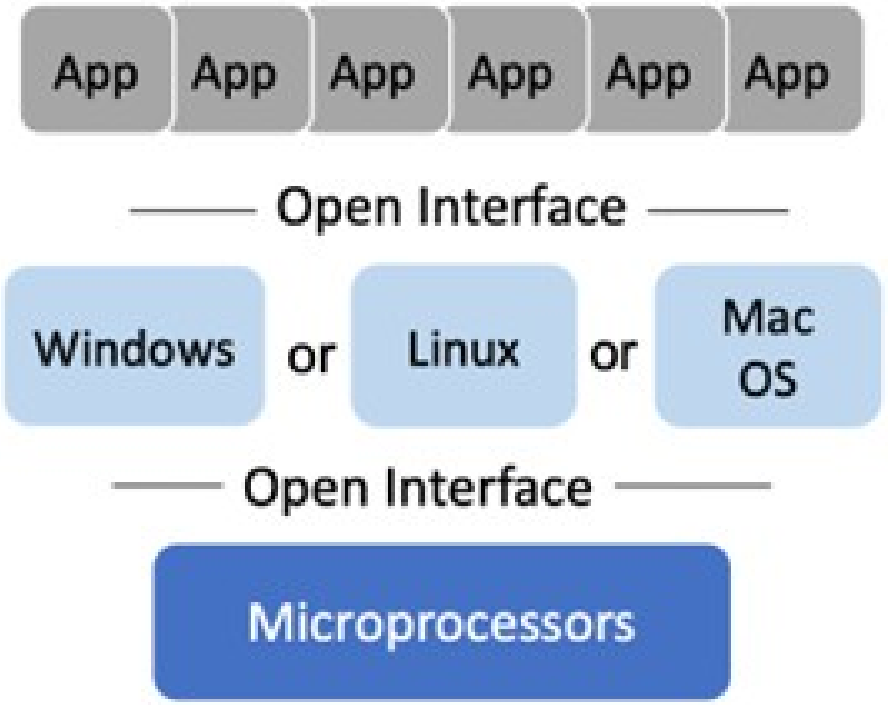
1. Advances in Microcomputer and microprocessor technology

- Transformation of the vertical mainframe market to a horizontal marketplace with open interfaces and multiple options available at every level.
- The introduction of powerful microprocessors (e.g., Intel x86 and Motorola 68000) and open source OS's (e.g., BSD Unix and Linux), helped transform that vertical market into a horizontal marketplace, with open interfaces spurring innovation at every level.









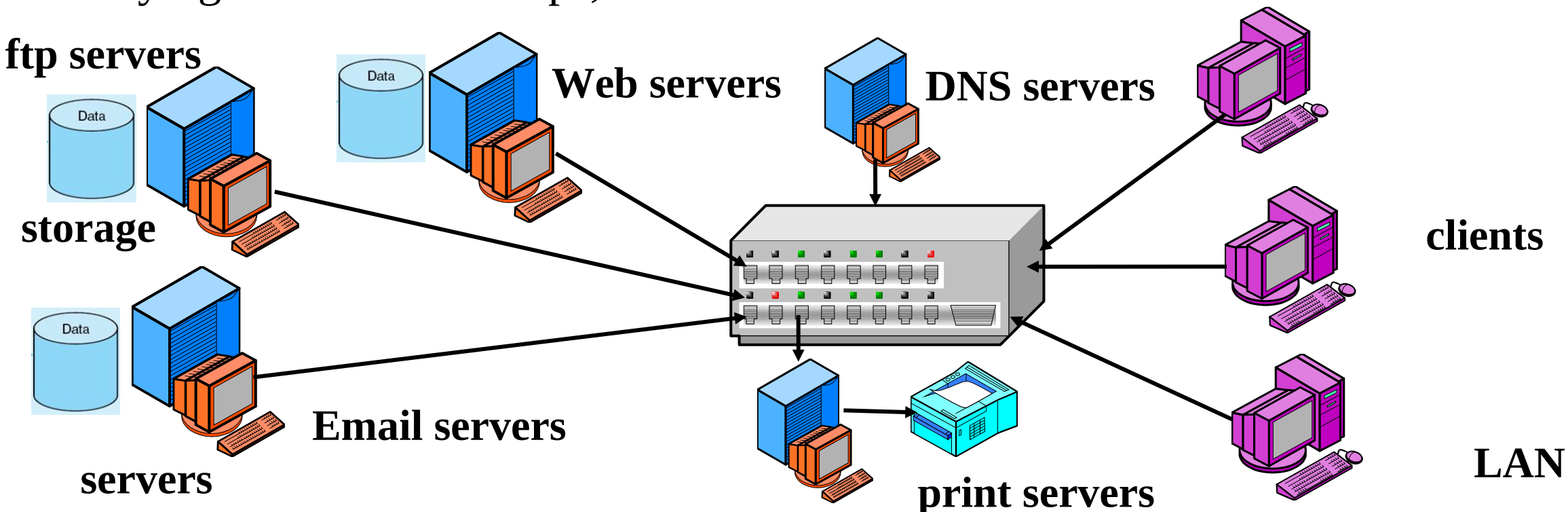
Clustering and virtualization

2. Evolution of Local Area Network LAN Technology

The second development was the invention of high-speed LAN and WAN computer networks.

LANs allow thousands of machines within a building or campus to be connected in such a way that small amounts of information can be transferred in a few microseconds or so. Larger amounts of data can be moved between machines at rates of 10^{12} bps.

WANs allow hundreds of millions of machines all over the earth to be connected at speeds varying from 10^4 to 10^9 bps, and sometimes even faster.



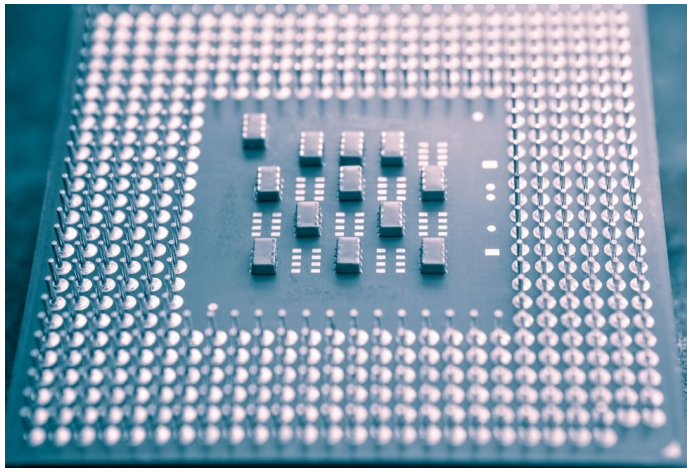
2. Evolution of Local Area Network LAN Technology

- ✓ A LAN consists of a number of different computers. Workstations WSs and PCs provide the front-end for network users. Different servers provide shared services.
- ✓ One or several network **file servers** provide data storage services. Any WS and PC may store files on disks maintained by these file servers.
- ✓ A **local name server** maps machine names to IP addresses, user names to user ids and group names to group ids. Any machine can request a service to resolve a certain name.
- ✓ One or **several print servers** control the access to shared printers. WSs and PCs have the server printing jobs for them.
- ✓ Another component provides a gateway to the WAN.
- ✓ Name servers provide the information regarding machine (by IP address), user names (by user ids), and group name (by group ids).
- ✓ LANs can be connected together to form Intranets.

3. PCs Technology

- Parallel to the development of increasingly powerful, reducing costs, and networked machines, we have also been able to witness miniaturization of computer systems with perhaps the **smartphone** as the most impressive outcome. Packed with powerful PCs, sensors, lots of memory, and a powerful CPU, these devices are nothing less than full-fledged computers.
- Along the same lines, **plug computers** and other so-called **Nano computers** are finding their way to the market.
- These small computers, often the size of a power adapter, can often be plugged directly into an outlet and offer near-desktop performance.





3. PCs Technology

A PC **desktop computers and notebook** can perform all of its input, processing, output, and storage activities by itself.

A **desktop** computer is designed so the system unit, input devices, output devices, and any other devices fit entirely on or under a desk or table. Variations of desktop computers include :

- Tower models (tall computers that can sit vertically on the floor),
- All-in-one computers (computers that combine monitor and system unit into a single device),
- Workstations (more expensive and powerful computers designed for work that requires intense calculation and graphics capabilities).

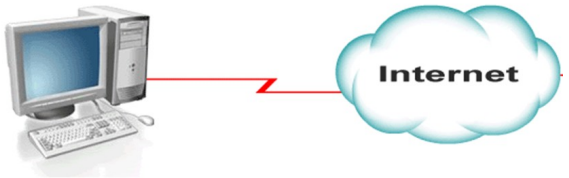
A **notebook** computer is a portable PC small enough fit on lap.

Notebook and desktop computers are used at home or in the office to perform application software-related tasks or to access the Internet.

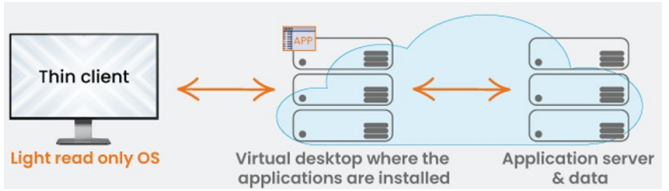
Personal Computers

Dump Terminals
Emulation Mode
Diskless PC

Standalone PC



Thin Client



Thick Client

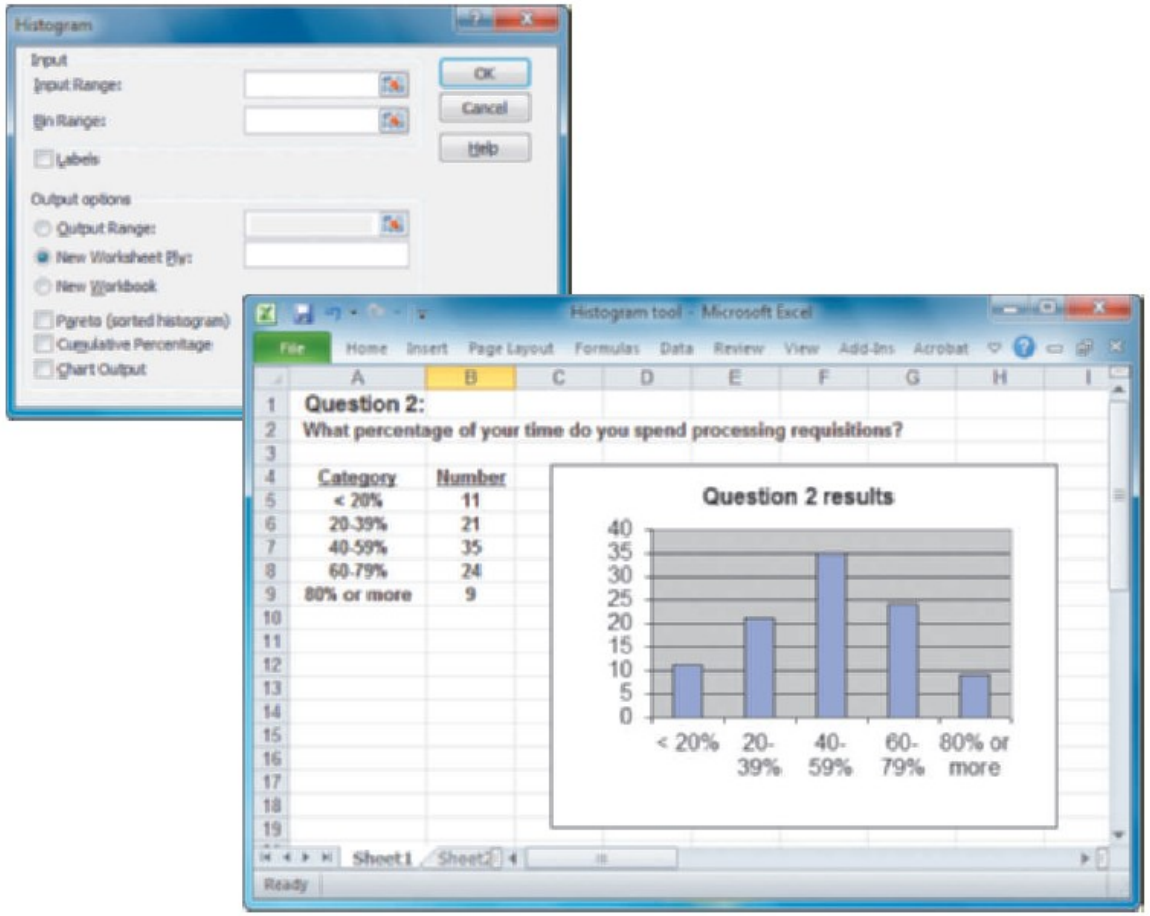


4. Need for multimedia for decision making

decision makers are asking for multimedia (text, audio, video, graphs) for decision making.

This requirement replace of text terminals with PCs and use of powerful servers.





Need for multimedia for decision making



The technologies that move the computer system systems from centralized to distributed

- The result of these technologies put together a computing system composed of many networked computers, be they large or small.
- These computers are generally geographically dispersed, for which reason they are usually said to form a **distributed system**.
- The size of a DCS vary from a handful of devices, to millions of computers.
- The interconnection network may be wired, wireless, or a combination of both.
- DCS are highly dynamic, in the sense that computers can join and leave, with the topology and performance of the underlying network almost continuously changing.

Summary for IT Motivations for Distributed Systems

Feature	Notes
Technological Changes	● **Growth in microelectronics** (e.g., VLSI – **V**ery **L**arge **S**cale **I**ntegration, etc) has enabled price-performance ratio to favor multiple low-performance processors rather than single high-performance processors. ● **Interconnection** and **communication costs** have fallen dramatically – **LANs**, etc are now readily available and cost-effective. ● **PCs and mobiles** are powerful, low cost devices, multifunction's, to replace terminals
User Needs	● Increased desire to provide computing power (processing) where it is needed. Users need more **sophisticated multimedia reports, facilities – faster, reliable, etc.** ● There is also managerial desire to Distribute the Computing. ● Managers need multimedia data for better decisions-making

The technologies that move the computer system systems from centralized to distributed

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DCS Definition(s)

Definition 1: "A DCS **runs on a collection of computers that do not have shared internal memory**, **yet looks like a single computer to its users**."

Definition 2: "A DCS is one in which hardware and software components located at **networked** computers communicate and coordinate their actions only by **message passing**."

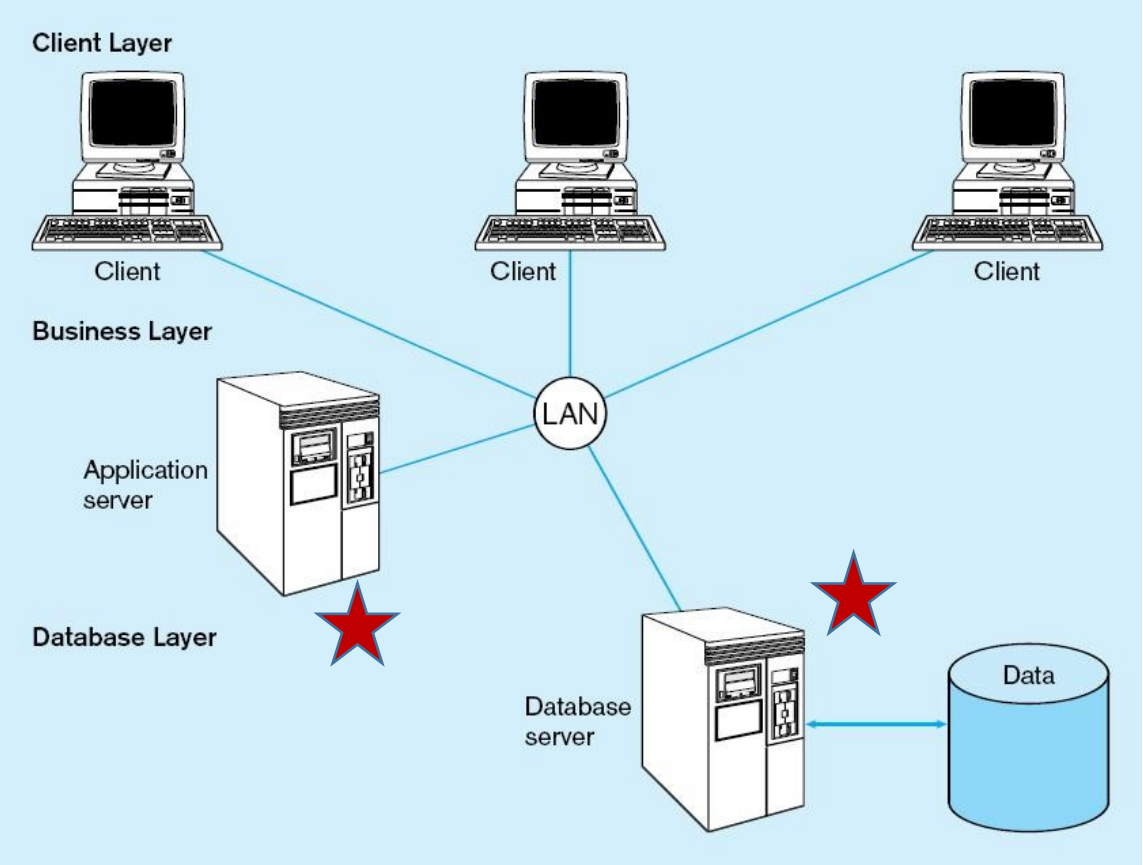
Definition 3: "A DCS is a collection of **independent** computers **that appear** to the users of the system as a single computer."

DCS Definition(s)

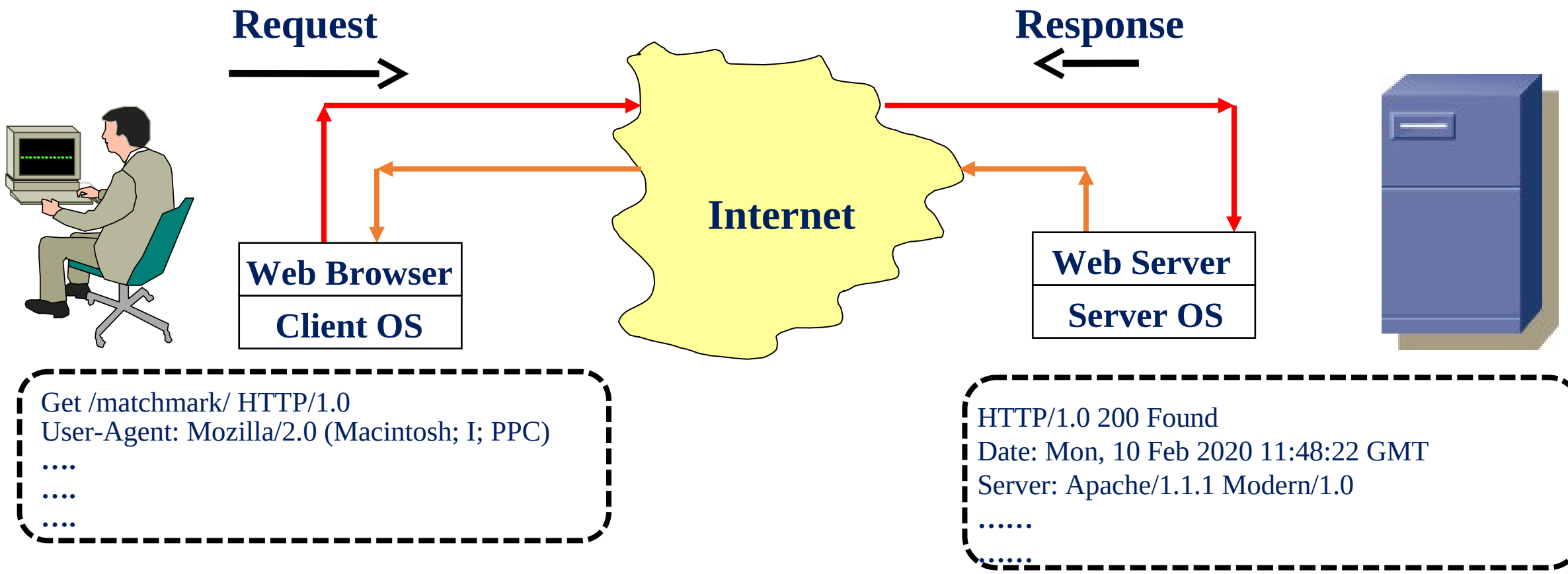
Definition 4: A DCS is one in which several autonomous processors and data stores supporting processes and/or databases interact in order to cooperate to achieve an overall goal.
The processes coordinate their activities and exchange information by means of information transferred over a communications network.

EXAMPLE OF DCS: LAN

- A LAN consists of a number of different computers.
- WSs and PCs provide the front-end for users.
- Different servers provide shared services and applications.
- One or several network file servers provide data storage services.
- Any WS and PC may store files on disks maintained by these file servers.
- Programmers could design programs that need application and database servers without any serious modifications on the LAN DCSs that appear to be the same as non-DCSs.
- **Achieving a single-system view is considered virtually**.



Basic Distribution Concept ….



Basic Distribution Concept ….

A **computing system** lives to execute **applications:**

Stand-alone application: **– Calculator/Word/game/……**

Network application: **– Email/Web/data and programs sharing**

Needed

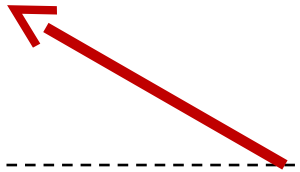
- **Hardware –** processors;
- **Control –** application and system software;
- **Data –** application and system information

Can these be distributed?

What gains and challenges with distribution?

Basic Distribution Concept: Onslow's Model

Hardware, Control, Data



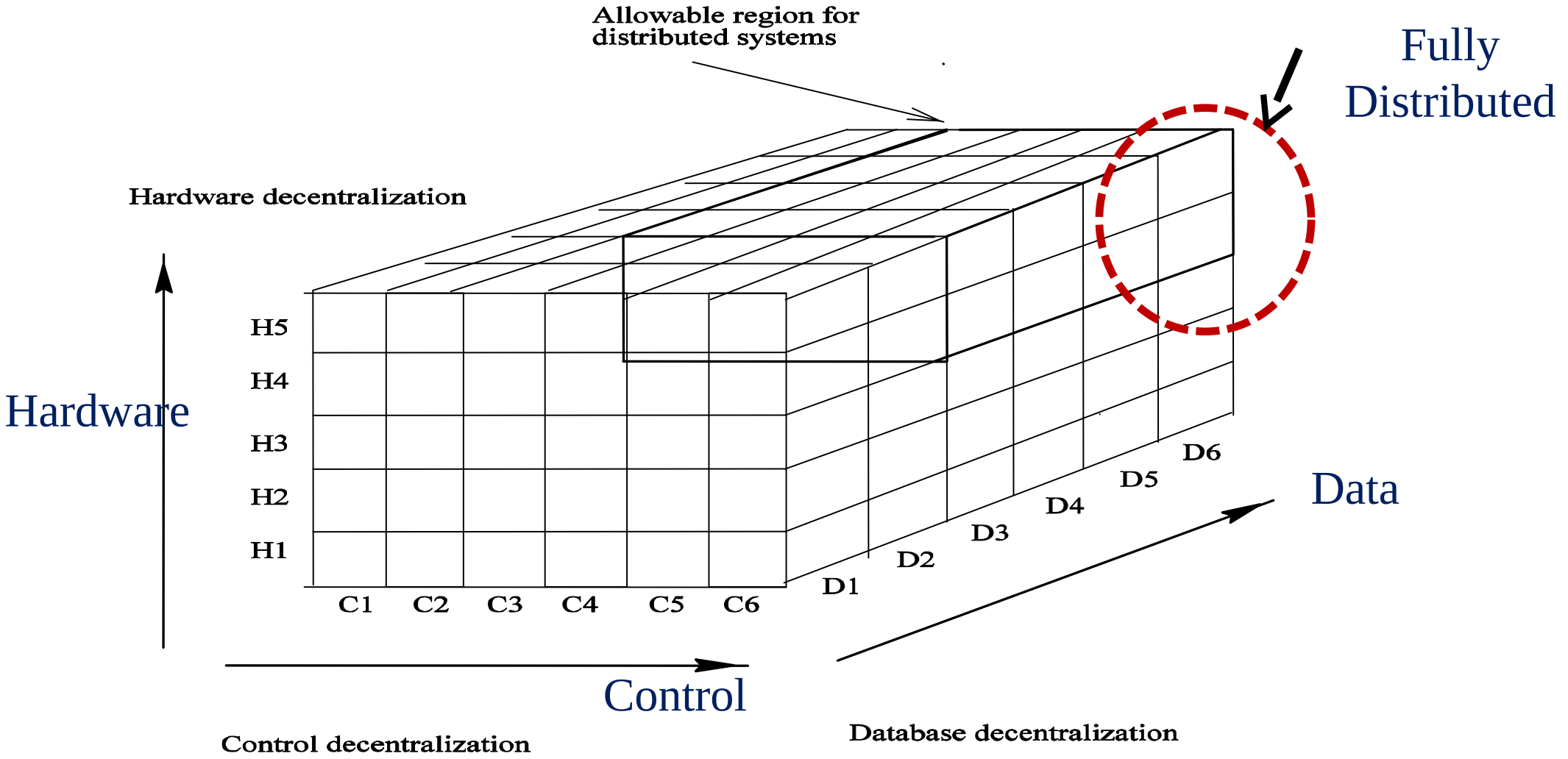
A Distributed System = **distributed hardware**
and/or **distributed control**
and/or **distributed data**

Distributed system = distributed hardware + distributed control +distributed data

Thus, a system could be classified as a distributed system if **all three categories (hardware, control, data)** reach a certain degree of decentralization.

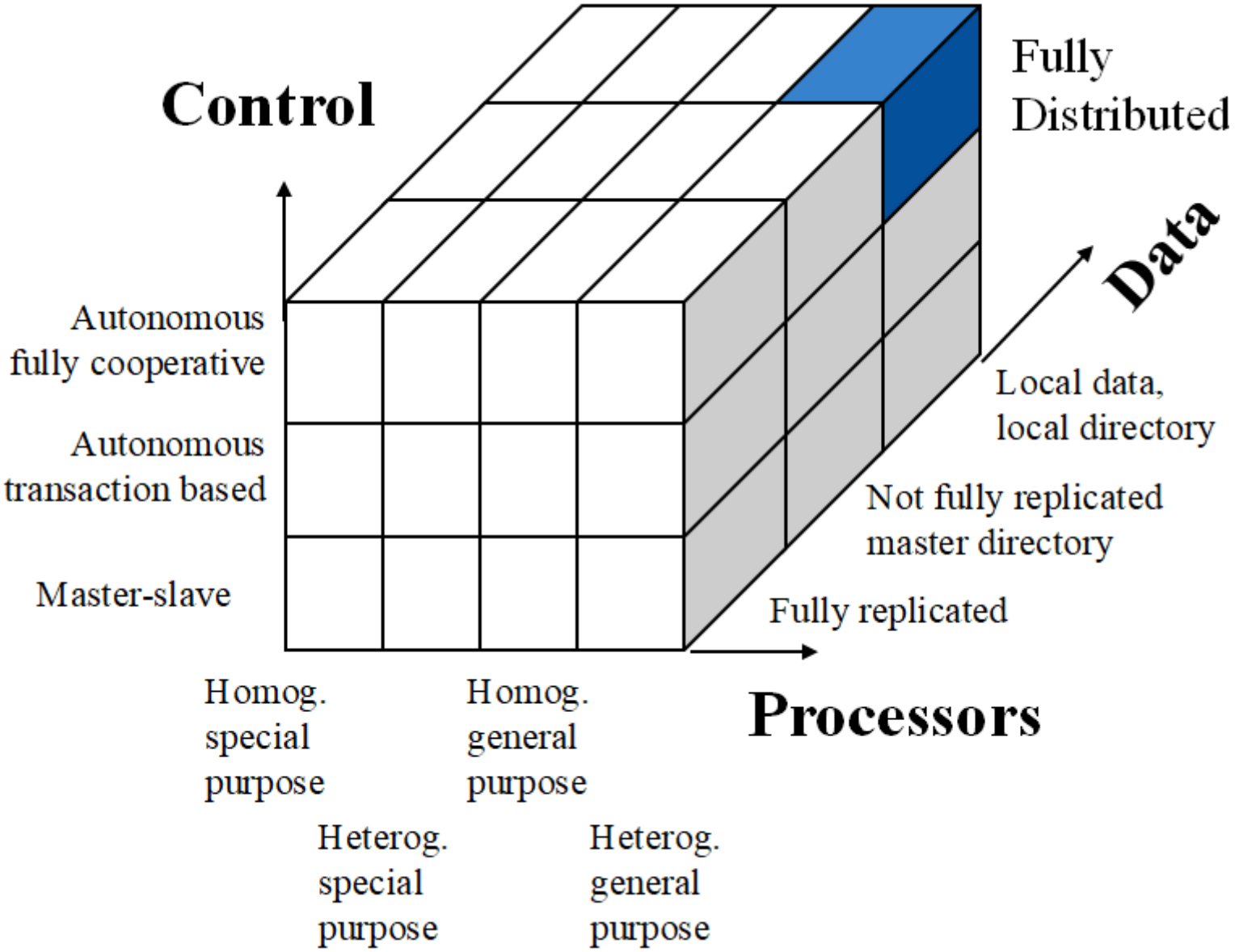
Onslow's Model for DCSs

Distributed system = distributed hardware + distributed control + distributed data
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Onslow's Model for DCSs

- Distributed system = distributed hardware + distributed control + distributed data
- A system could be classified as a distributed system if **all three categories (hardware, control, data)** reach a certain degree of decentralization.



Onslow's Model for DCSs

CONTROL

- Single fixed control point.
- Single dynamic control point. In multiple CPU cases the controller changes from time to time among CPUs.
- A fixed master/slave structure. For example, in a system with one CPU and one co-processor, the CPU is a fixed master and the co-processor is a fixed slave.
- A dynamic master/slave structure. The role of master/slave is modifiable by software.
- Multiple homogeneous control points where copies of the same controller are used.
- Multiple heterogeneous control points where different controllers are used.

DATA

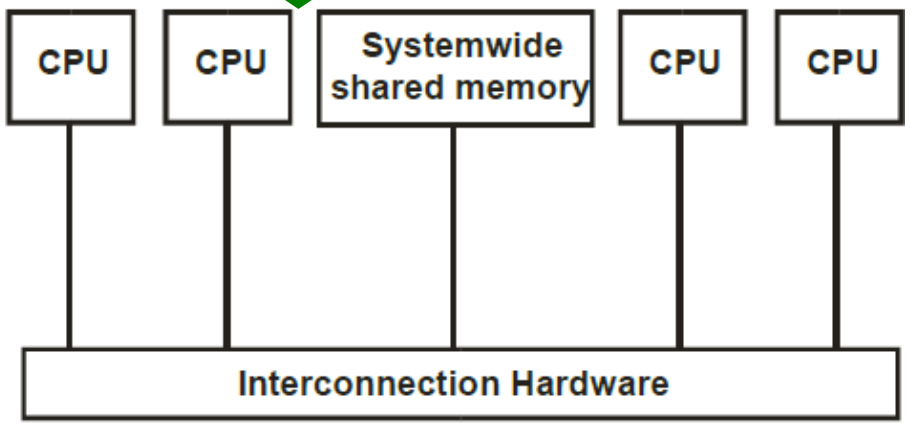
- Centralized databases with a single copy of both files and directory.
- Distributed files with a single centralized directory and no local directory.
- Replicated database with a copy of files and a directory at each site.
- Partitioned database with a master that keeps a complete duplicate copy of all files.
- Partitioned database with a master that keeps only a complete directory.
- Partitioned database with no master file or directory.

Homogenous Vs Heterogeneous Computer Systems

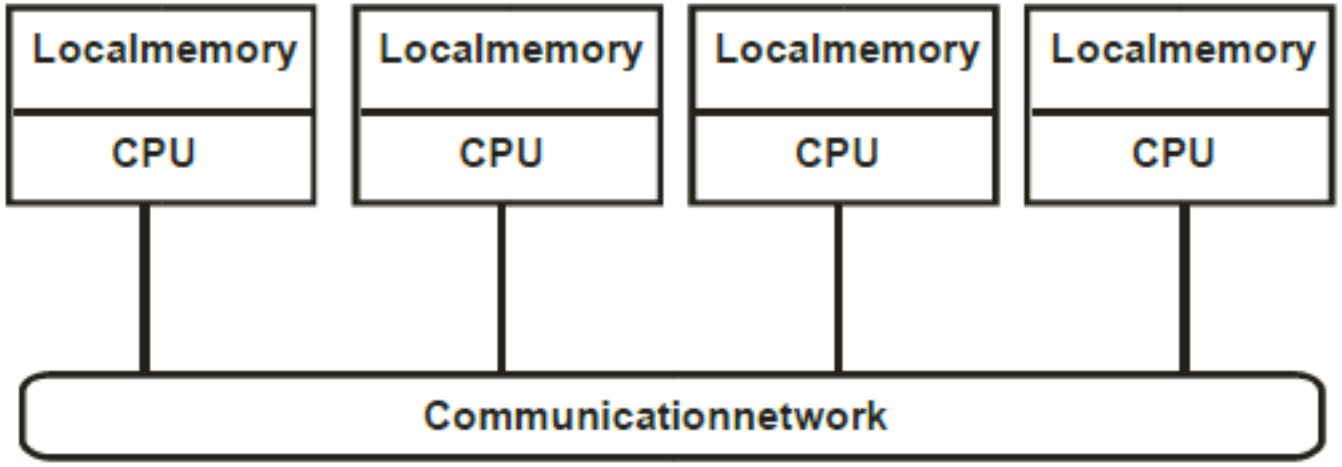
Subsystems	Homogenous Systems	Heterogeneous Systems
Computer hardware	Same	Variant/Different
Operating systems	Same	Variant/Different
Programming languages	Same	Variant/Different
Implementations by different developers/vendors	Same	Variant/Different
Networks	Same	Variant/Different
Design and Management	Simple	Difficult
Growth	Simple	Difficult
Increased Performance	Simple	Difficult

Multiple Computing Types

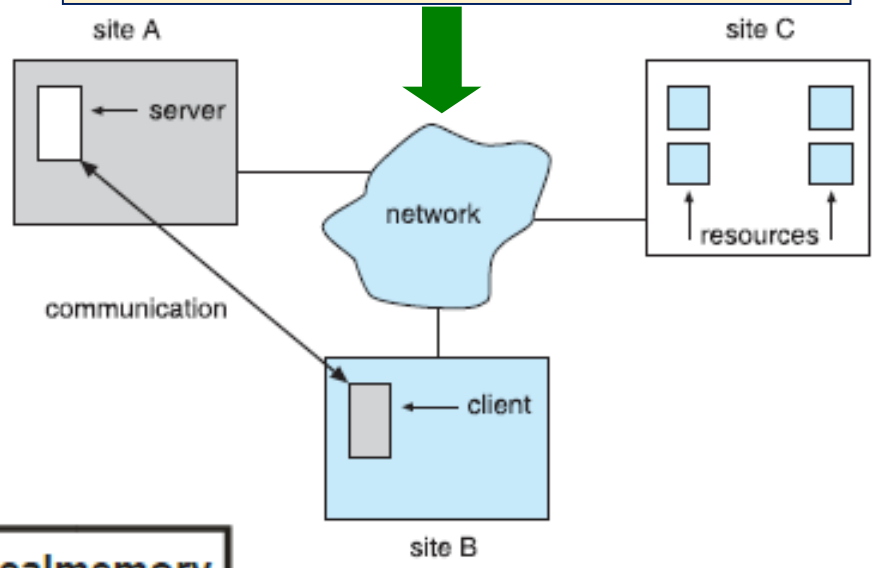
Tightly coupled systems: SHARED MEMORY



Loosely Coupled Systems: PRIVATE MEMORY



Networked Systems: PRIVATE AND NETWORKED MEMORIES



Tightly coupled systems:

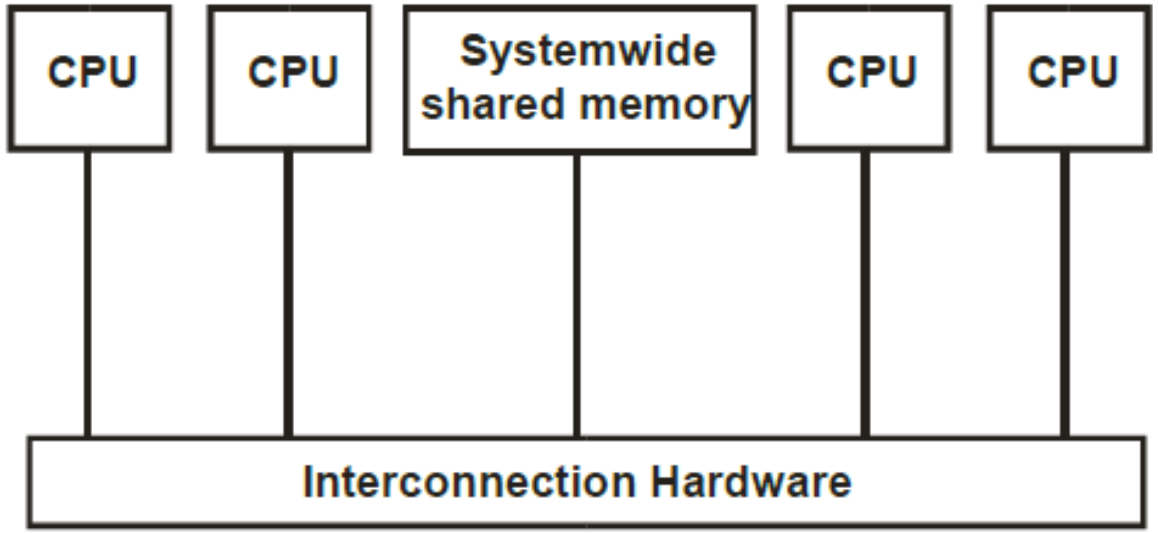
Tightly coupled systems are multiprocessor systems. They are a population of processors sharing addressing to memory and devices. This sharing is accomplished by utilizing single OS.

Tightly coupled systems are referred to as parallel processing systems or shared memory, were a single system wide primary memory (address space) is shared by all the processors.

If any processor writes, for example, the value 100 to the memory location x, any other processor subsequently reading from location x will get the value 100.

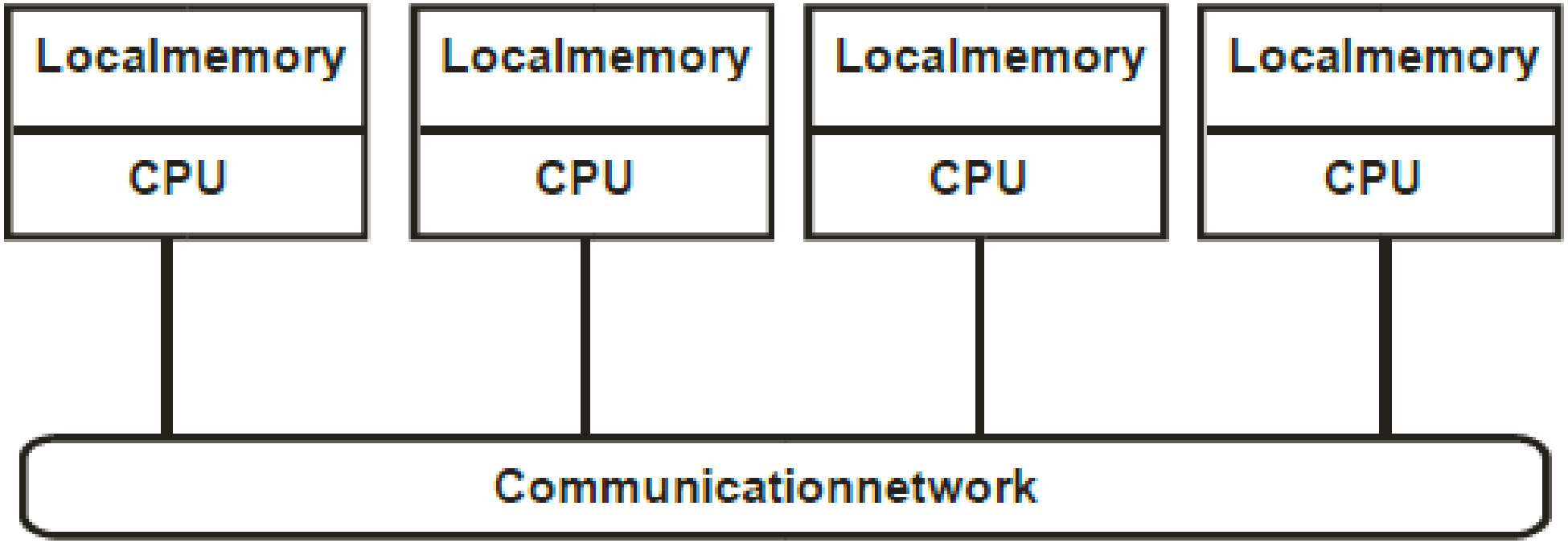
In these systems, any communication between the processors takes place by the shared memory.

In tightly coupled systems, the number of processors that can be deployed is small and limited by the bandwidth of the shared memory.



Loosely Coupled Systems: private memory

- The loose interconnection of computer systems comes very close to the concept of distributed processing by the attribute of multiple complete systems.
- They are systems connected over I/O pathways traditionally distinct from networks.
- Many of these pathways differ from tightly coupled systems only in the physical sense that they do not share memory or operating systems.
- They may be functionally quite dependent, and as a result take on the characteristics of a single node, a single 'system' rather than an interconnection of two nodes.

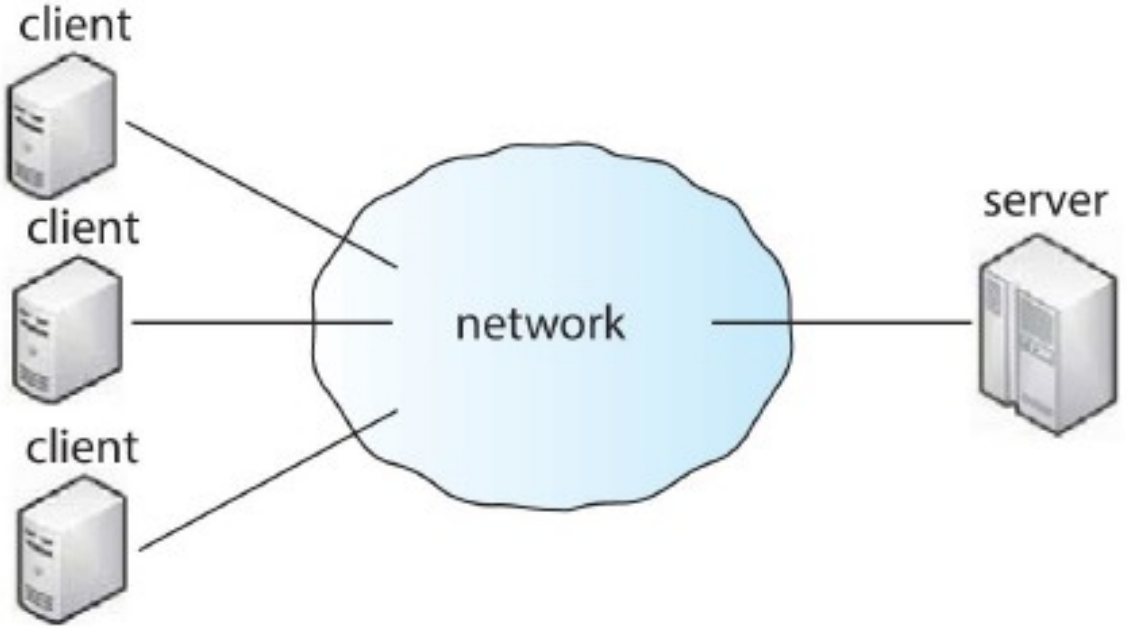


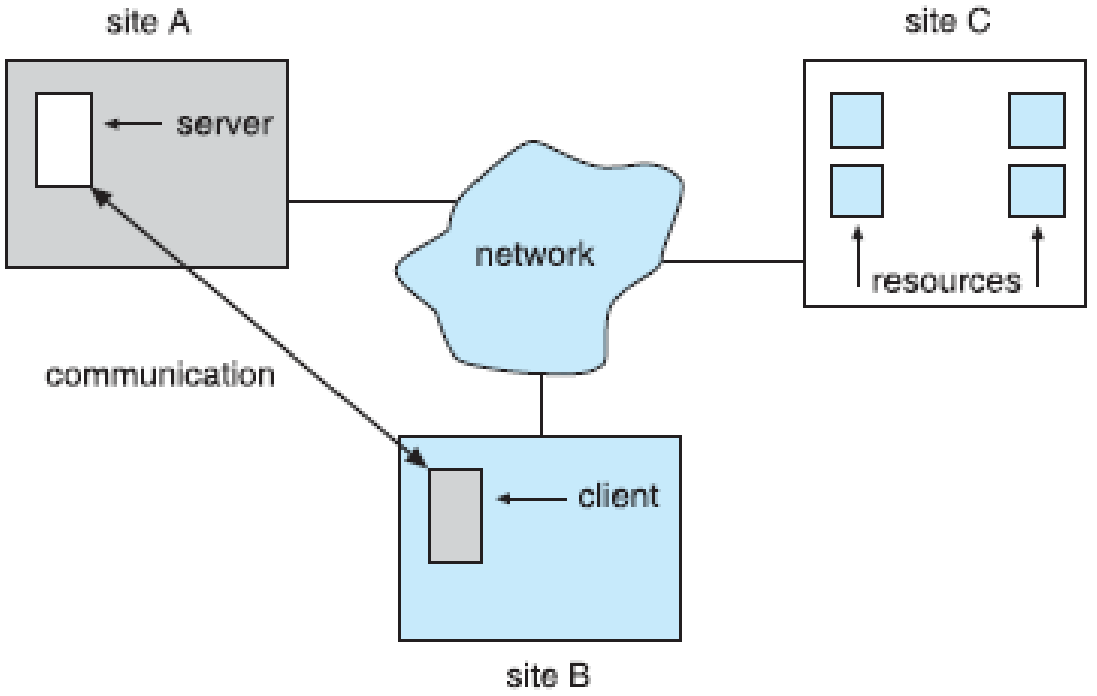
Difference between Loosely Coupled and Tightly Coupled Multiprocessor System

S.O	Loosely Coupled	Tightly Coupled
1.	There is distributed memory in loosely coupled multiprocessor system.	There is shared memory, in tightly coupled multiprocessor system.
2.	Loosely Coupled Multiprocessor System has low data rate.	Tightly coupled multiprocessor system has high data rate.
3.	In loosely coupled multiprocessor, Memory conflicts don't take place.	While tightly coupled multiprocessor system have memory conflicts.
4.	Loosely Coupled Multiprocessor system has low degree of interaction between tasks.	Tightly Coupled multiprocessor system has high degree of interaction between tasks.
5.	In loosely coupled multiprocessor, there is direct connection between processor and I/O devices.	While in tightly coupled multiprocessor, IOPIN helps connection between processor and I/O devices.

Networked Systems: private memory and networked storage

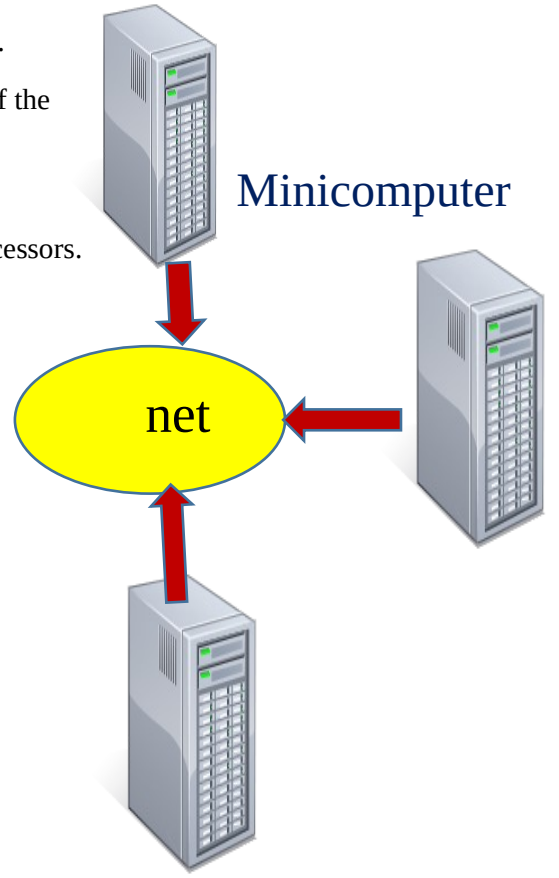
- There are many types of networks displaying a broad range of characteristics.
- They can vary from completely autonomous systems with the added ability to communicate electronically, to diskless workstations completely dependent on the network server for all secondary storage and communications operations.
- Each type of network has some unique set of properties that makes it attractive based on service provided, cost, reliability, serviceability, flexibility, etc.
- As well, each unique network possesses some set of properties that make it similar in nature to other network types.





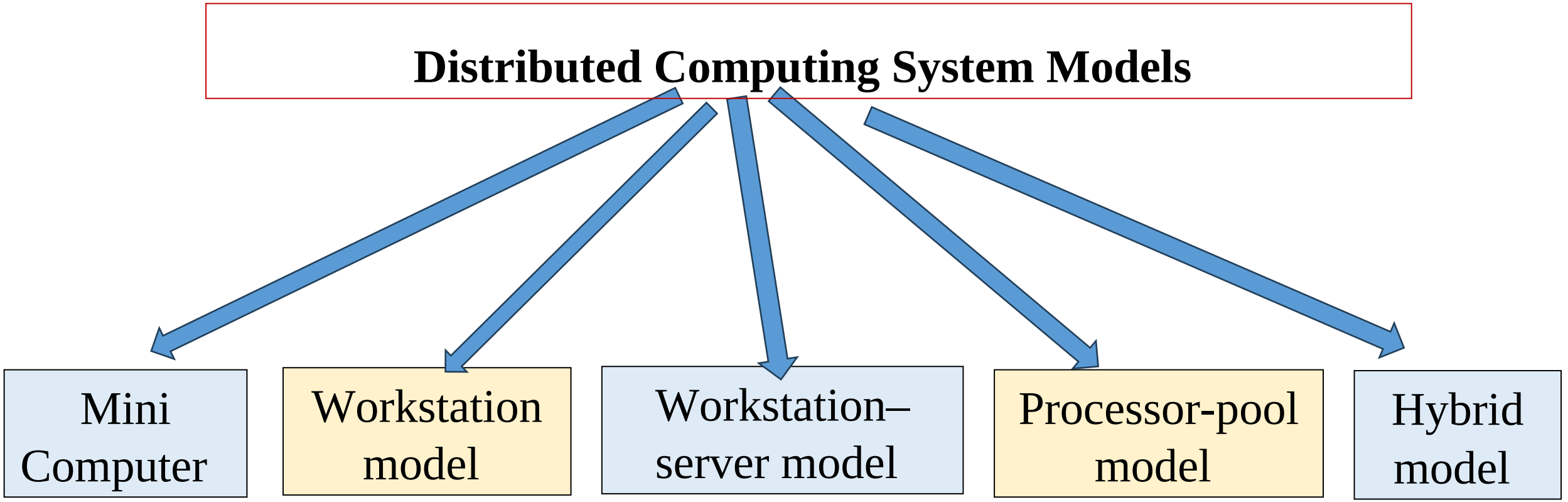
Networked Systems

- loosely coupled systems using networks are referred to as DCS. In these systems, the processors do not share memory, and each processor has its own local memory.
- If a processor writes the value 100 to the memory location x, this write operation will only change the contents of its local memory and will not affect the contents of the memory.
- All physical communication between the processors is done by passing messages across the interconnection network.
- Processors can be located far from each other to cover a wider geographical area, DCS are more freely expandable and can have an almost unlimited number of processors.



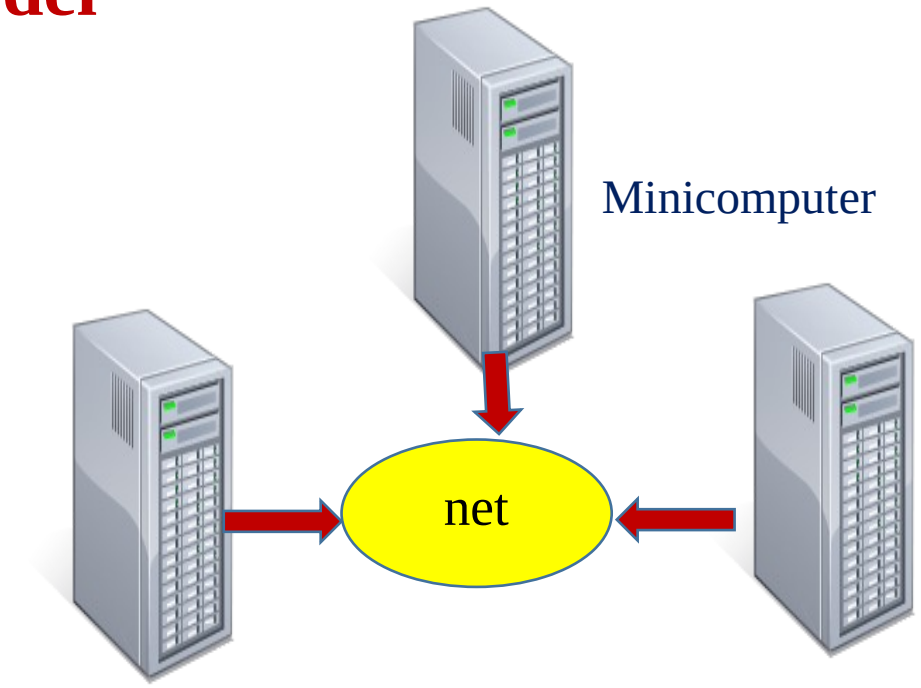
- DCS is basically a collection of processors interconnected by a communication network in which each processor has its own local memory and other peripherals, and the communication between any two processors of the system takes place by message passing over the communication network.
- For a particular processor, its own resources are local, whereas the other processors and their resources are remote. A processor and its resources are referred to as a node or site or machine of the DCS.

Distributed Computing System Models



Minicomputer model

- The minicomputer model is a simple extension of the centralized time-sharing system.
- A DCS consists of a few minicomputers interconnected by a communication network where each minicomputer usually has multiple users simultaneously logged on to it.
- Several interactive terminals are connected to each minicomputer. Each user logged on to one specific minicomputer has remote access to other minicomputers.
- The network allows a user to access remote resources that are available on some machine other than the one on to which the user is currently logged.
- The minicomputer model may be used when resource sharing with remote users is desire d.
- **Early Internet is example of mini computer DCS.**

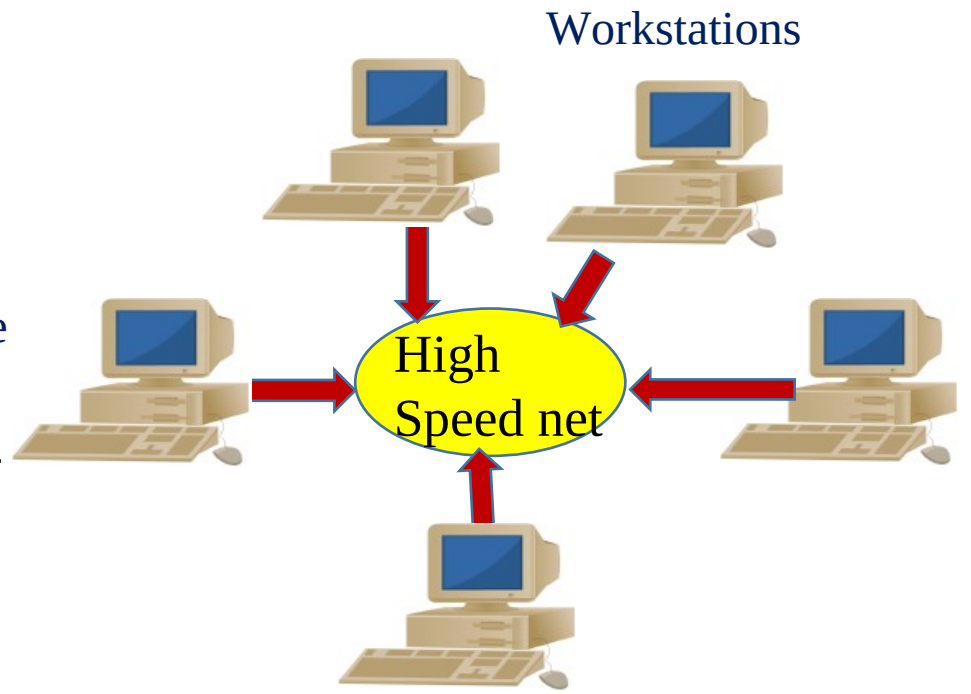


Drawbacks of Minicomputer Model:

- Each user is logged on to one specific minicomputer, with remote access to other minicomputers.
- The processing of tasks are done by the processors of a single minicomputer.
- This model is very expensive.

2.Workstation WS Model

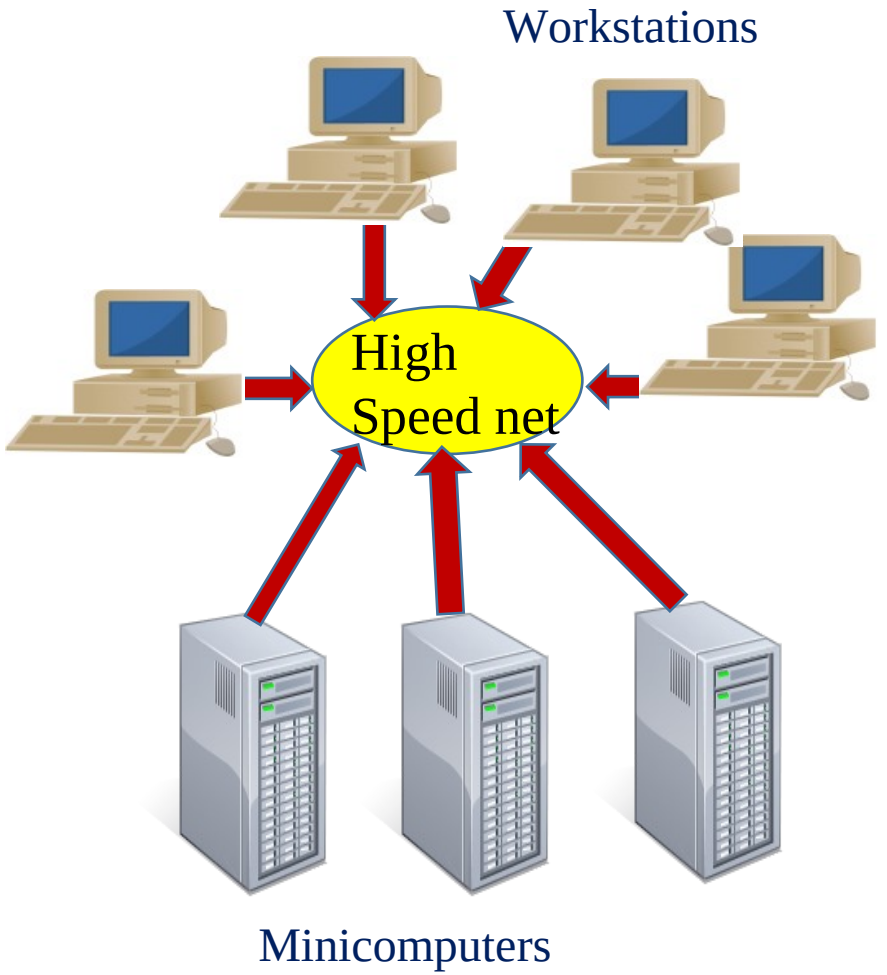
- **WS Model** Consists of network of powerful PCs, Each one with its own hard disk and local filesystem, Interconnected over high speed network
- A DCS based on the WS model consists of several WSs interconnected by a communication network.
- An organization may have several WSs located throughout an infrastructure where each WS is equipped with its own disk & serves as a single-user computer.
- In such an environment, at any one time a significant proportion of the WSs are idle which results in the waste of large amounts of CPU time.
- Therefore, the idea of the WS model is to interconnect all these WSs by a high-speed LAN so that idle WSs used to process jobs of users who are logged onto other WSs & do not have sufficient processing power at their own WSs to get their jobs processed efficiently.
- Example: Sprite system & Xerox PARC



3. Workstation-Server Model

- **WS-Server Model** Consists of multiple workstations coupled with powerful servers with extra hardware to store the file systems and other software like databases

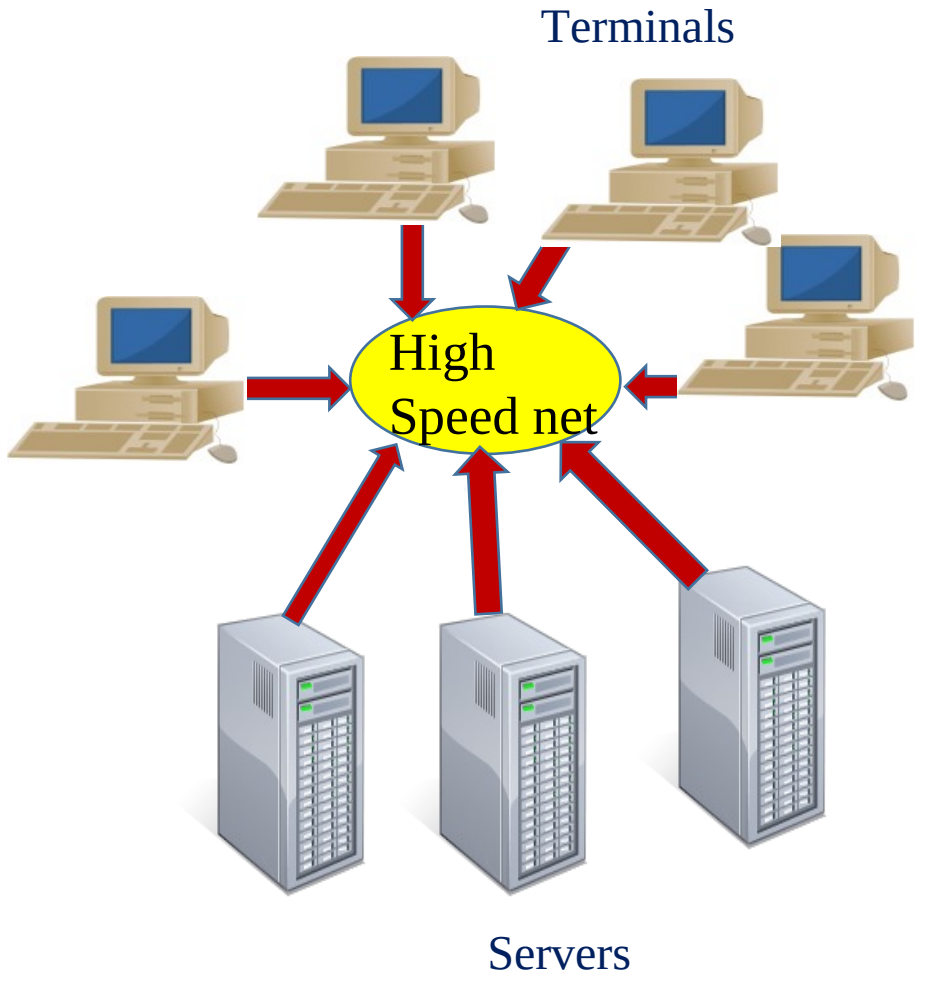
  - The WS model is a network of **DISK-FULL PC WS** having its own disk & a local file system.
  - A DCS based on the WS-server model consists of a few minicomputers and several WSs interconnected by a network.
  - A user logs onto a WS called his own PC.
  - Normal computation activities required by the user's processes are performed at the user's PC, but requests for services provided by special servers are sent to a server providing that type of service that performs the user's requested activity & returns the result of request processing to the user's workstation.
  - The user's processes need not migrated to the server machines for getting the work done by those machines.



4. Processor-pool Model

- Consists of multiple processors: a pool of processors and a group of workstations

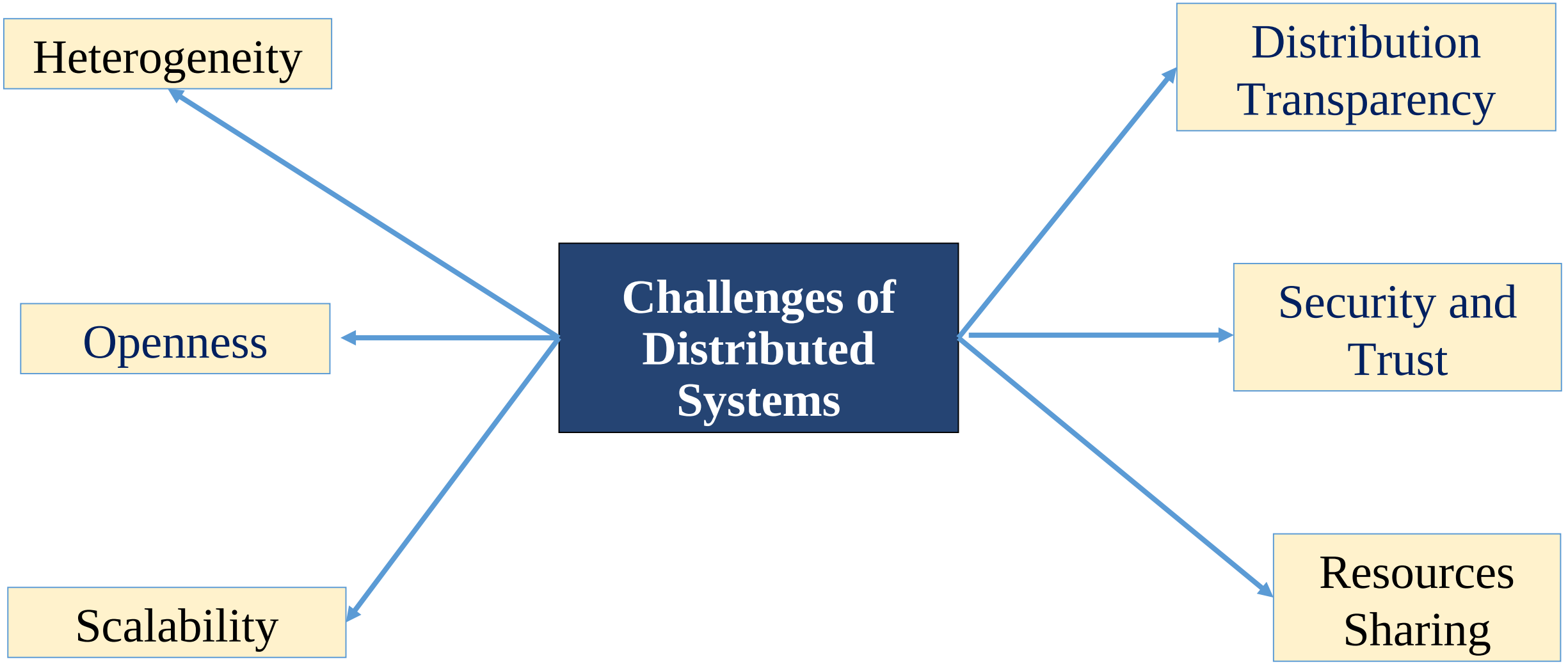
  - The processor-pool model is based on the observation that most of the time a user does not need any computing power but once in a while the user may need a very large amount of computing power for a short time.
  - Therefore, unlike the WS-SERVER model in which a processor is allocated to each user, in processor-pool model the processors are pooled together to be shared by the users as needed.
    - The pool of processors consists of a large number of microcomputers & minicomputers attached to the network.
    - Each processor in the pool has its own memory to load & run a system program or an application program of the distributed computing system.
  - In this model no home machine is present & the user does not log onto any machine.
  - This model has better utilization of processing power & greater flexibility.
    - Example: Cambridge Distributed Computing System.



5.Hybrid Model:

- The workstation-server model has a large number of computer users only performing simple interactive tasks &-executing small programs.
- In a working environment that has groups of users who often perform jobs needing massive computation, the processor-pool model is more attractive & suitable.
- To combine Advantages of workstation-server & processor-pool models, a hybrid model can be used to build a distributed system.
- The processors in the pool can be allocated dynamically for computations that are too large or require several computers for execution.
- The hybrid model gives guaranteed response to interactive jobs allowing them to be more processed in local workstations of the users.

Challenges of Distributed Systems



Challenges of Distributed Systems

- As the scope and scale of distributed systems and applications are extended, different challenges are likely to be encountered. In this section, the foremost challenges are described:
- **Heterogeneity:** DCSs enable users to access services and run applications over a heterogeneous collection of computers and networks.
- Heterogeneity (that is, variety and difference) applies to all of the following: networks, computer hardware, operating systems, programming languages, and implementations by different developers.

- **Openness:** The openness of a computer system is the characteristic that determines whether the system can be extended and re-implemented in various ways.
- The openness of a DCS is determined primarily by the degree to which new resource-sharing services can be added and be made available for use by a variety of client programs.
- Openness cannot be achieved unless the specification and documentation of the key software interfaces of the components of a system are made available to software developers.

Challenges of Distributed Systems

- **Scalability:** Distributed systems operate effectively and efficiently at many different scales, ranging from a small intranet to the Internet.
- The number of computers and servers in the Internet has increased dramatically.
- A system is described as scalable if it will remain effective when there is a significant increase in the number of resources and the number of users.

- **Distribution Transparency:** Transparency is defined as the concealment from the user and the application programmer of the separation of components in a distributed system.
- Thus, the system is perceived as a whole rather than as a collection of independent components. The problem with striving for distribution transparency in very large systems is that performance will degrade to unacceptable level.
- Moreover, network latencies have a natural lower bound that becomes noticeable, when dealing with long-haul connections.

Challenges of Distributed Systems

- **Security and Trust:** Many of the information resources that are made available and maintained in distributed systems have a high intrinsic value to their users. Their security is therefore of considerable importance. The decentralized organization of distributed systems raises serious challenges in domains of security and trust management. Implementing a secure distributed system requires solutions that can efficiently address different security issues.

- **Resources Sharing:** Decentralized scheduler organization negates the limitations of centralized organization with respect to fault-tolerance, scalability, and autonomy.
- This approach scales well for both, a small scale resource sharing environment (e.g. Resource sharing under the same administrative domain) in a large scale environment.
- However, this approach raises serious challenges in the domain of distributed information management, enforcing system wide coordination, security, resource user authenticity, and resource provider’s policy heterogeneity.
- There are many distributed systems in operation today.